2026 | Version 1.0

Clinic Management System

Complete Business & Technical Analysis

BRD | SRS | 15 UML Diagrams | 7 Analytical Tables

PROJECT HIGHLIGHTS

15 UML Diagrams | 7 Analytical Tables | 30 FR + 33 NFR

Prepared by

Abdullah Ibrahim Elsayed Mohamed Tawfik

Systems Analyst | Information Systems Graduate

iabdalla885@gmail.com | https://www.linkedin.com/in/abdullah-ibrahim-a70152422

Document Control

Item	Detail
Document Title	Clinic Management System — Business & Systems Analysis
Version	1.0
Author	[Abdullah Ibrahim Elsayed Mohamed Tawfik] — Systems Analyst
Audience	Hiring Managers, Lead Analysts, Solution Architects, Healthcare IT Recruiters
Purpose	Portfolio Case Study demonstrating end-to-end systems analysis for a healthcare domain
Confidentiality	Public (portfolio) — no real patient data used
Status	Final — Approved for portfolio distribution

Revision History

Version	Date	Author	Change
0.1	Aug 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Initial business analysis — problem statement & business objectives
0.5	Sep 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Added BRD, SRS, and 15 UML diagrams
1.0	Oct 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Final — complete BRD/SRS + 7 analytical tables + user stories

This document follows a standard revision workflow (Draft → Internal Review → Baseline) consistent with enterprise documentation practices in regulated industries.

1. Executive Summary

A mid-sized private clinic in Cairo, Egypt with 12 doctors, 8 receptionists, and 180 daily patients operates entirely on paper. This document presents a complete analysis and design for a Clinic Management System that will digitize all operations.

1.1 The Problem

- Average patient waiting time of 45 minutes
- Loss of 3 to 5 patient files per week
- Billing errors in 8% of invoices
- No real-time management visibility
- 25% no-show rate due to lack of reminders

1.2 Business Objectives

ID	Objective	Current	Target
BO-01	Patient wait time	45 min	Under 15 min
BO-02	Lost records	3-5/week	Zero
BO-03	Billing accuracy	92%	99%
BO-04	Report generation	2 days	Under 5 min
BO-05	Patient satisfaction	6.2	Above 8.5
BO-06	Administrative cost	Baseline	Reduce 25%

2. Business Requirements Document

ID	Requirement	Priority	Rationale
BR-01	Online appointment booking	High	Reduce phone load
BR-02	Digital medical records	High	Eliminate paper
BR-03	Multiple user roles	High	Data privacy
BR-04	Auto-generated invoices	High	Reduce errors
BR-05	SMS and email notifications	Medium	Reduce no-shows
BR-06	Reports	Medium	Data-driven decisions
BR-07	Multiple payment methods	High	Convenience
BR-08	Audit trail	High	Compliance
BR-09	Arabic and English	Medium	Diverse patients
BR-10	Payment gateway	High	Card payments

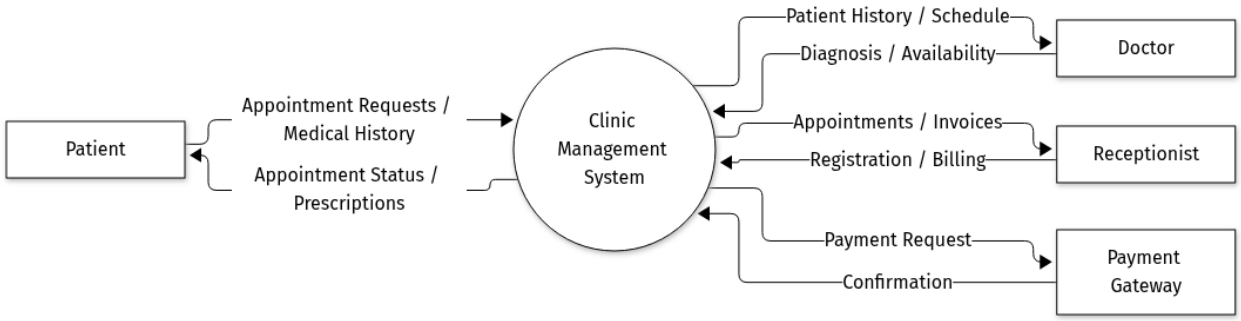
3. Software Requirements Specification

ID	Category	Requirement	Priority
FR-01	Patient	Register new patient	High
FR-02	Patient	Search patient	High
FR-03	Patient	Update profile	Medium
FR-04	Appointment	Display availability	High
FR-05	Appointment	Book appointment	High
FR-06	Appointment	Prevent double-booking	High
FR-07	Appointment	Reschedule	Medium
FR-08	Appointment	Cancel	Medium
FR-09	Records	View history	High
FR-10	Records	Add diagnosis	High
FR-11	Records	Prescriptions	High
FR-12	Billing	Auto-invoice	High
FR-13	Billing	Payment methods	High
FR-14	Billing	Receipts	Medium
FR-15	Security	Authentication	High

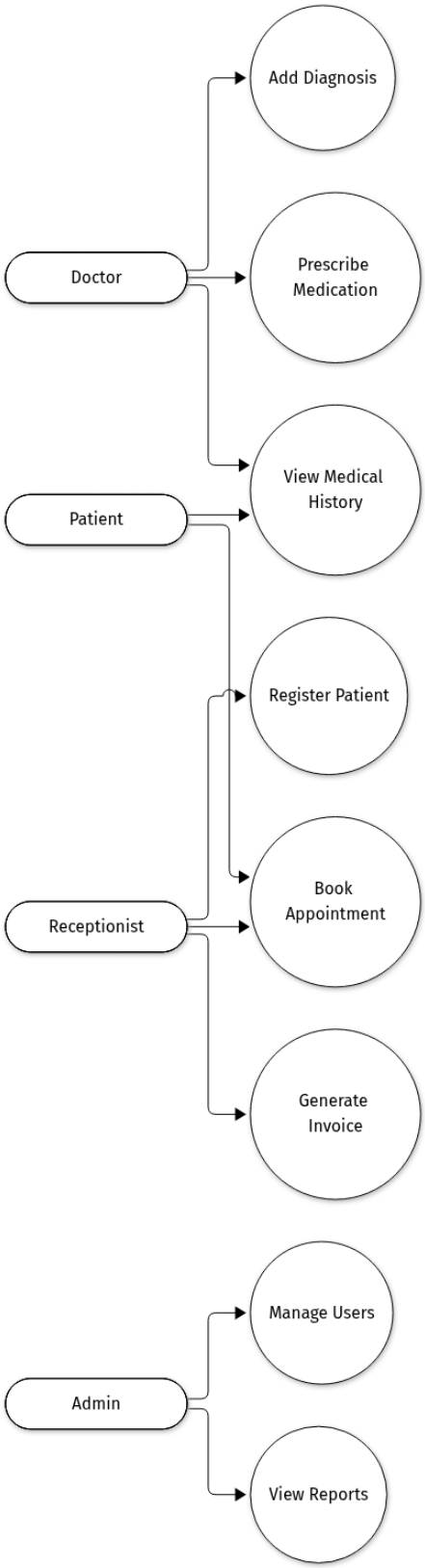
4. System Diagrams

This section contains 15 diagrams covering all aspects of the system.

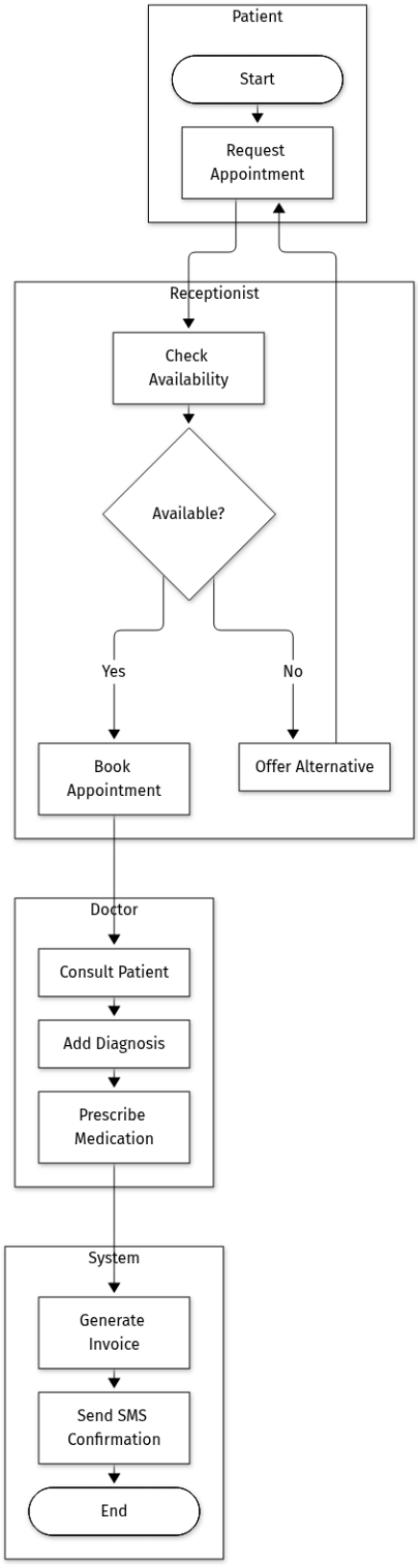
4.1 Context Diagram (DFD Level 0)



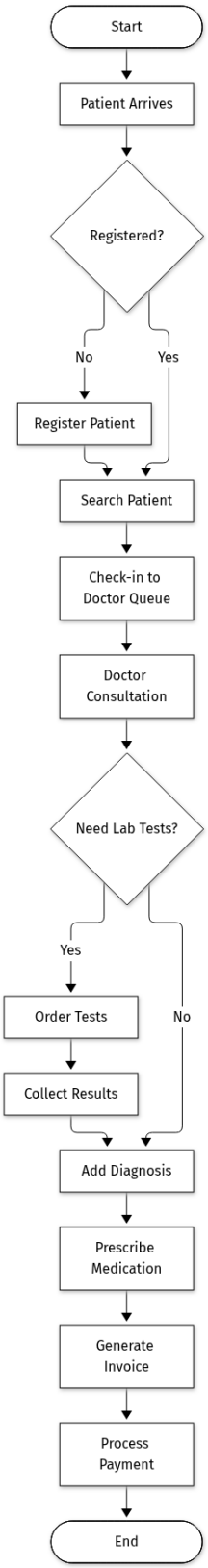
4.2 Use Case Diagram



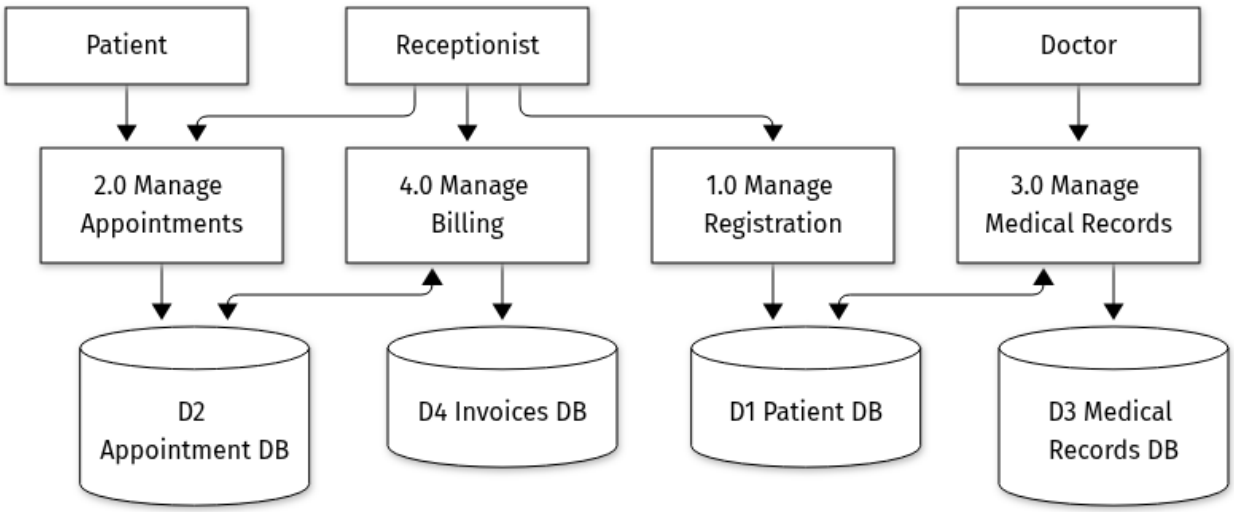
4.3 BPMN Diagram



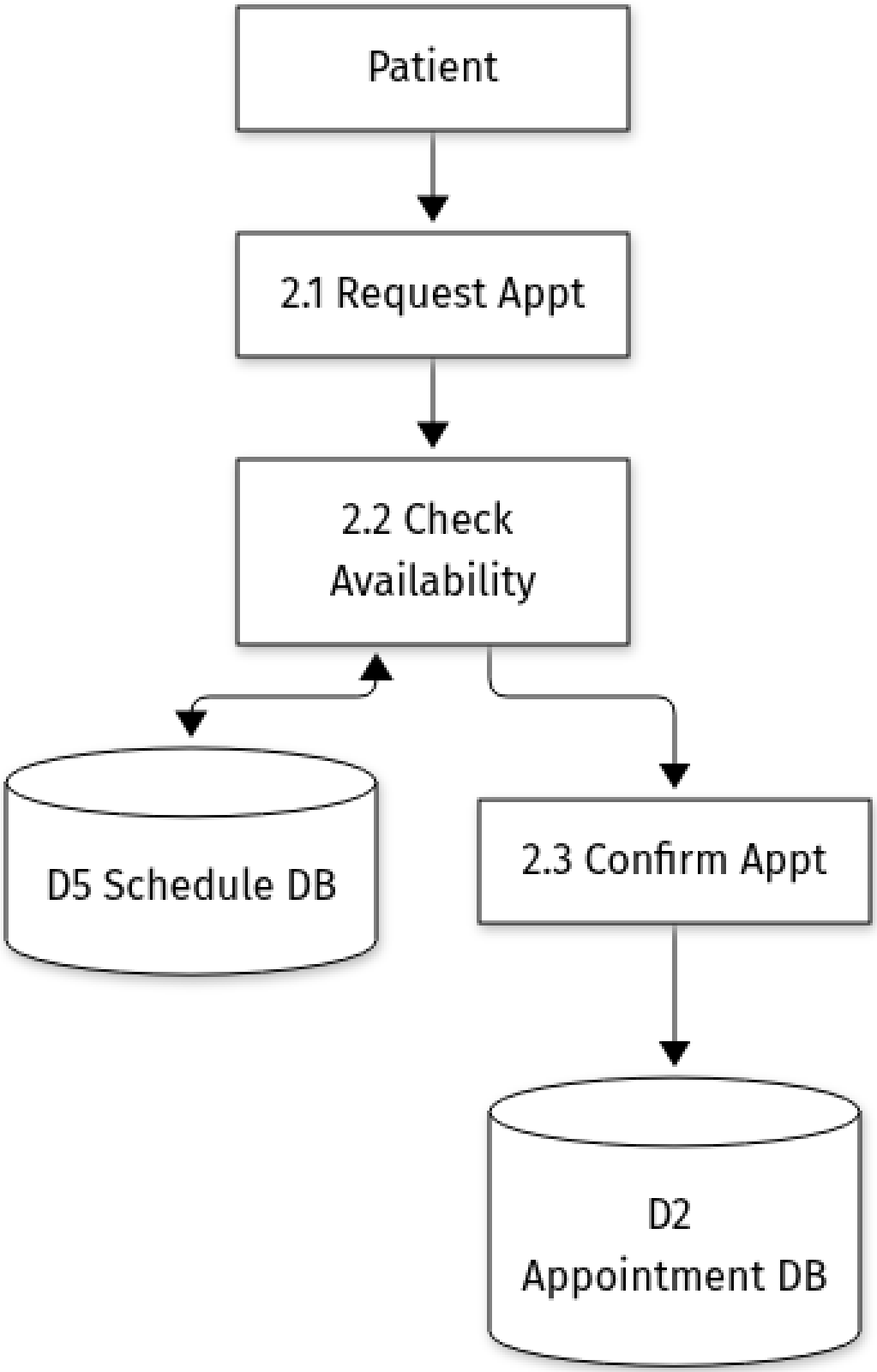
4.4 Activity Diagram



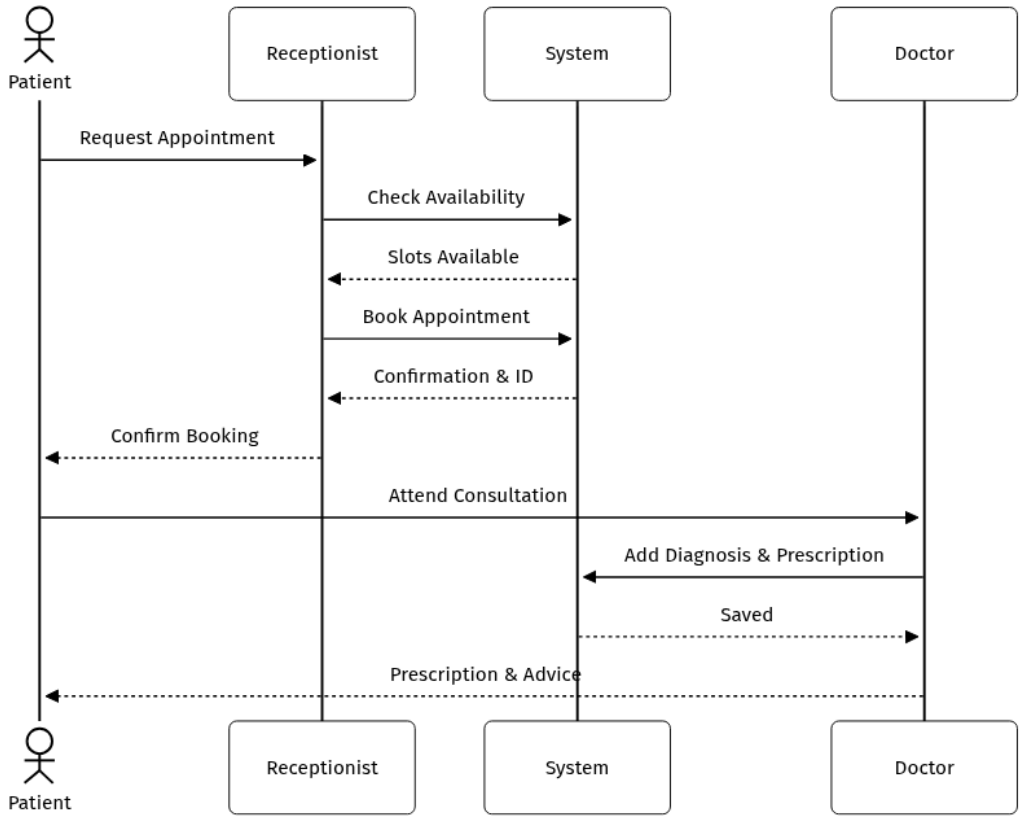
4.5 DFD Level 1



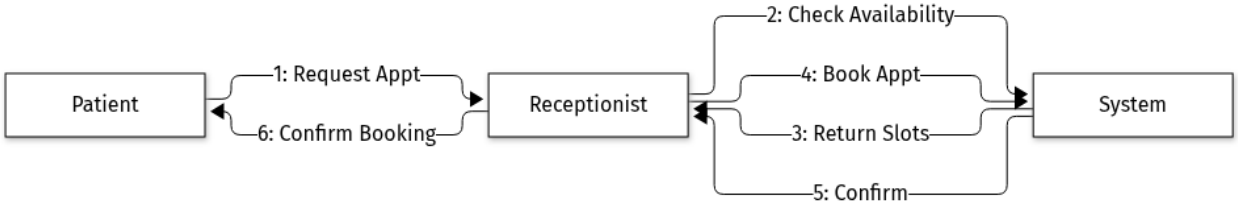
4.6 DFD Level 2



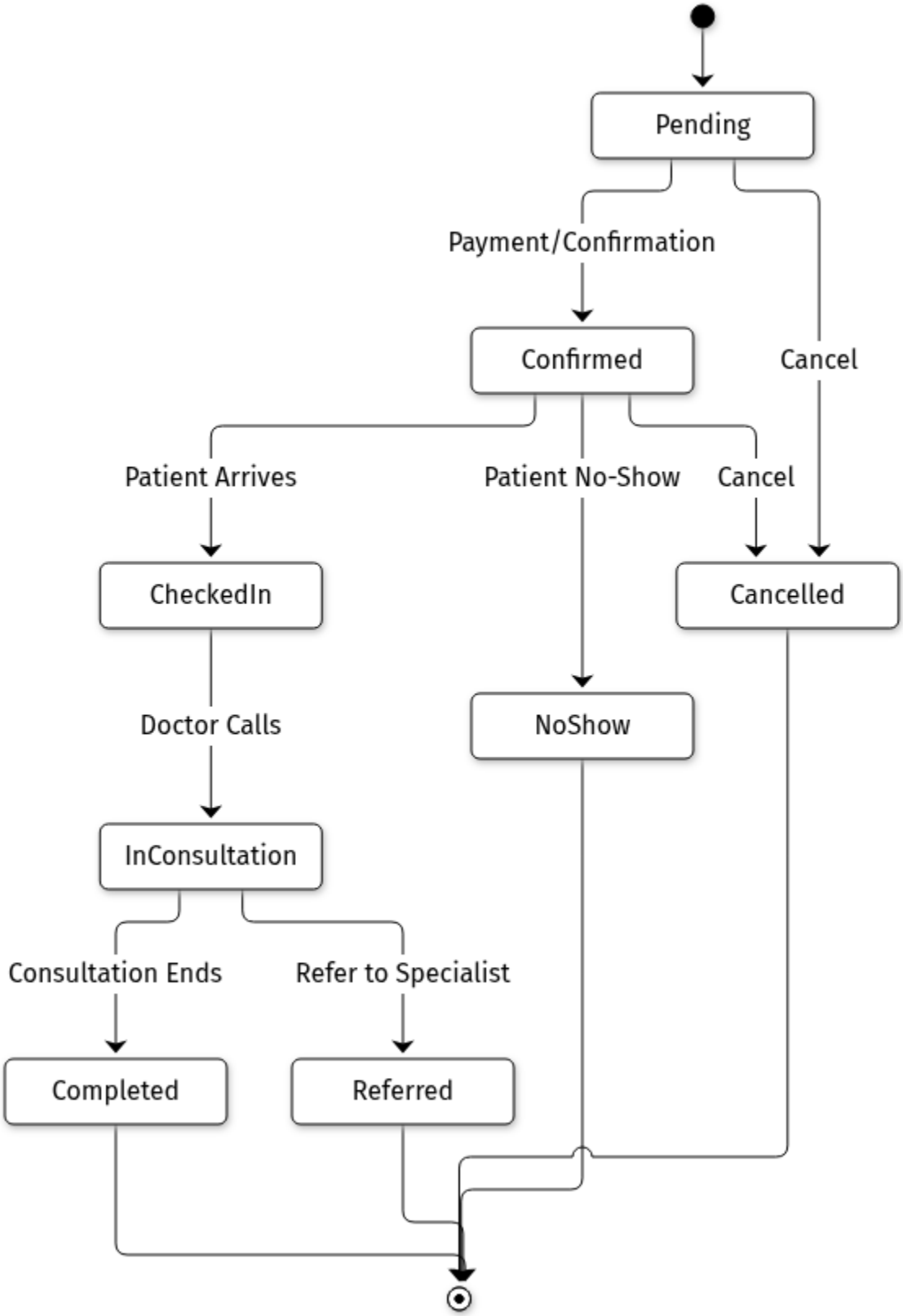
4.7 Sequence Diagram



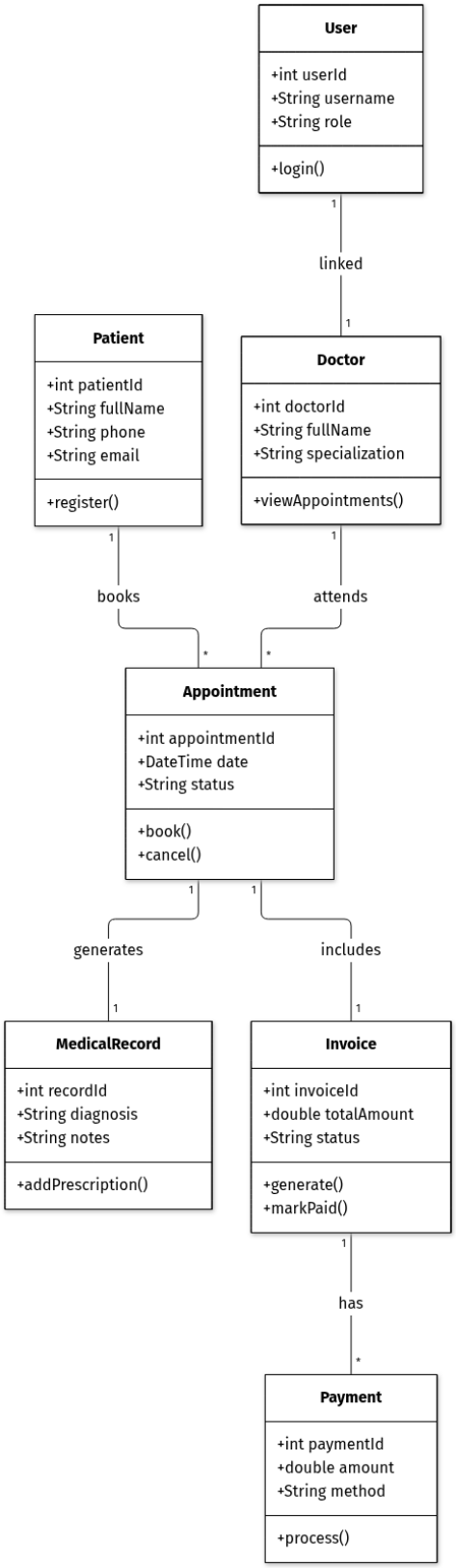
4.8 Communication Diagram



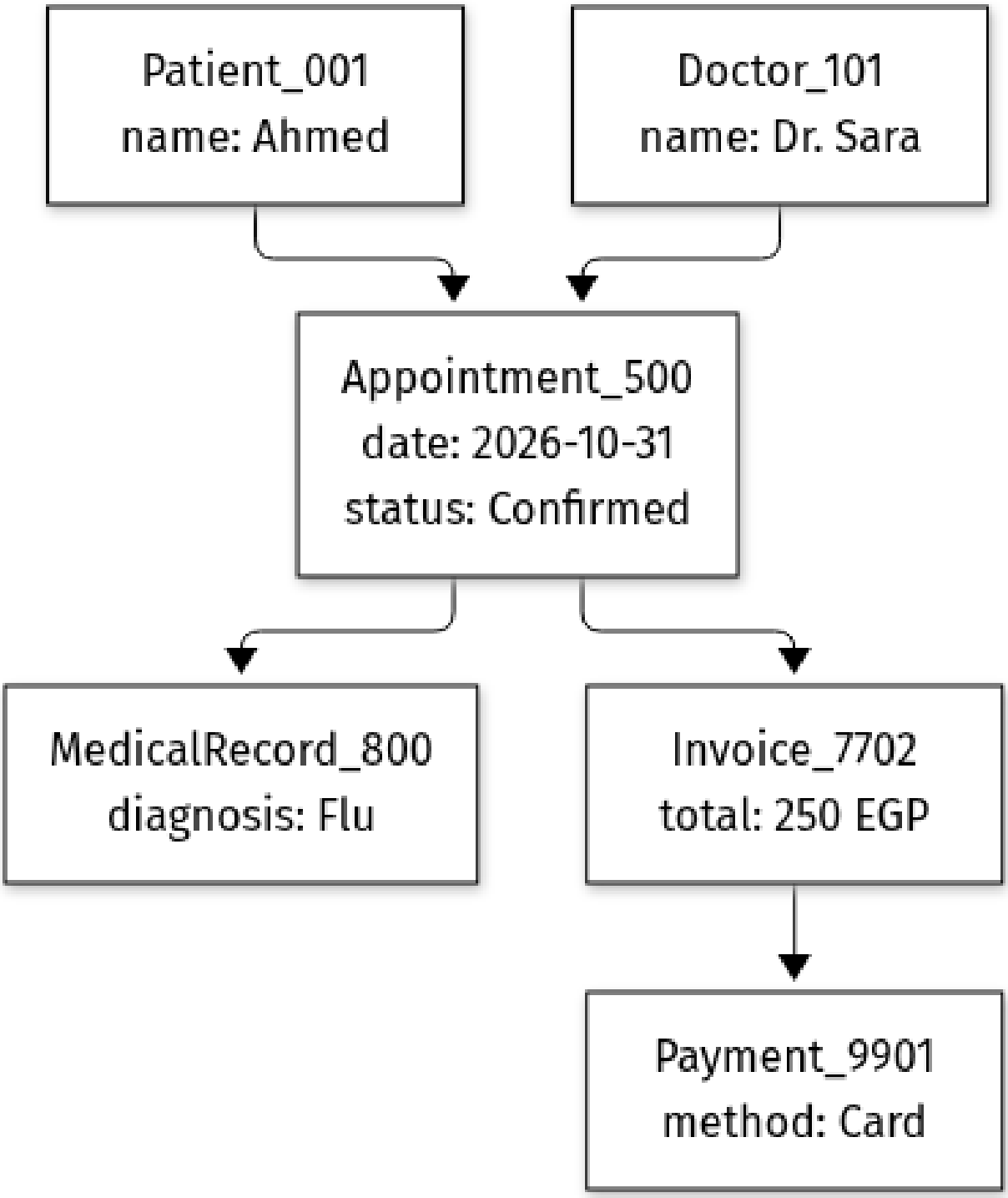
4.9 State Diagram



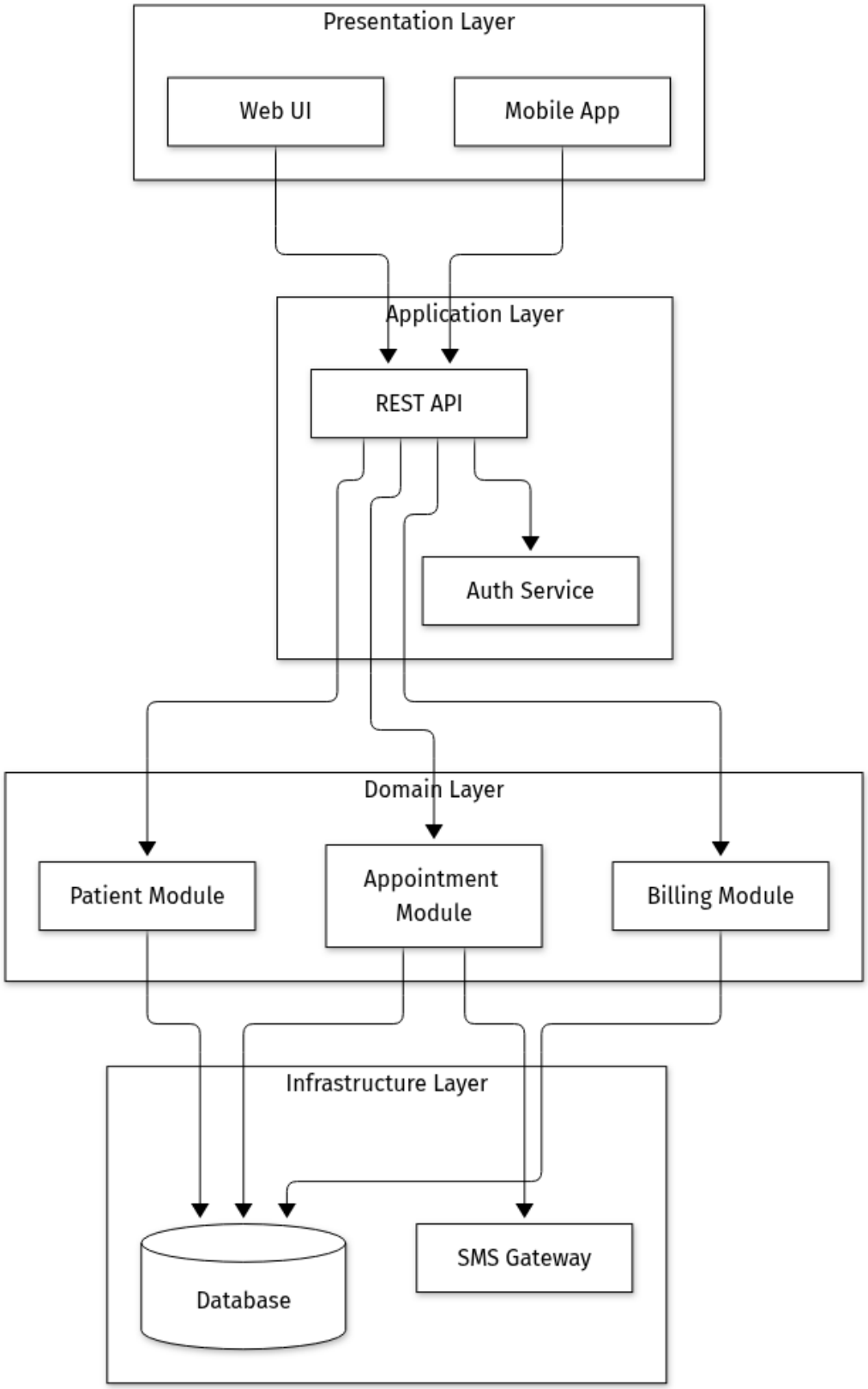
4.10 Class Diagram



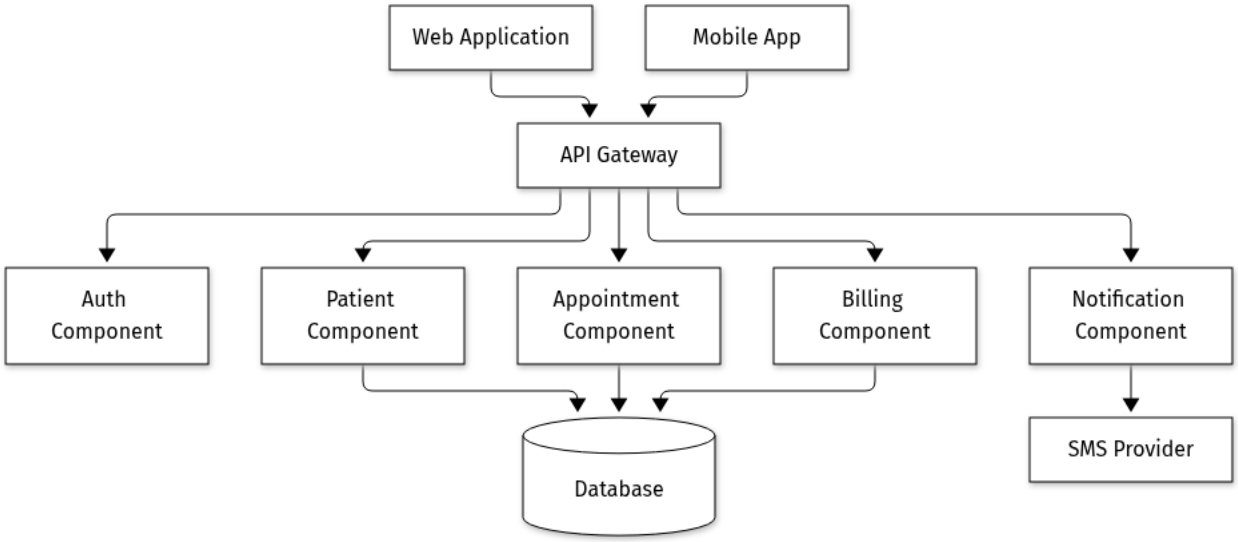
4.11 Object Diagram



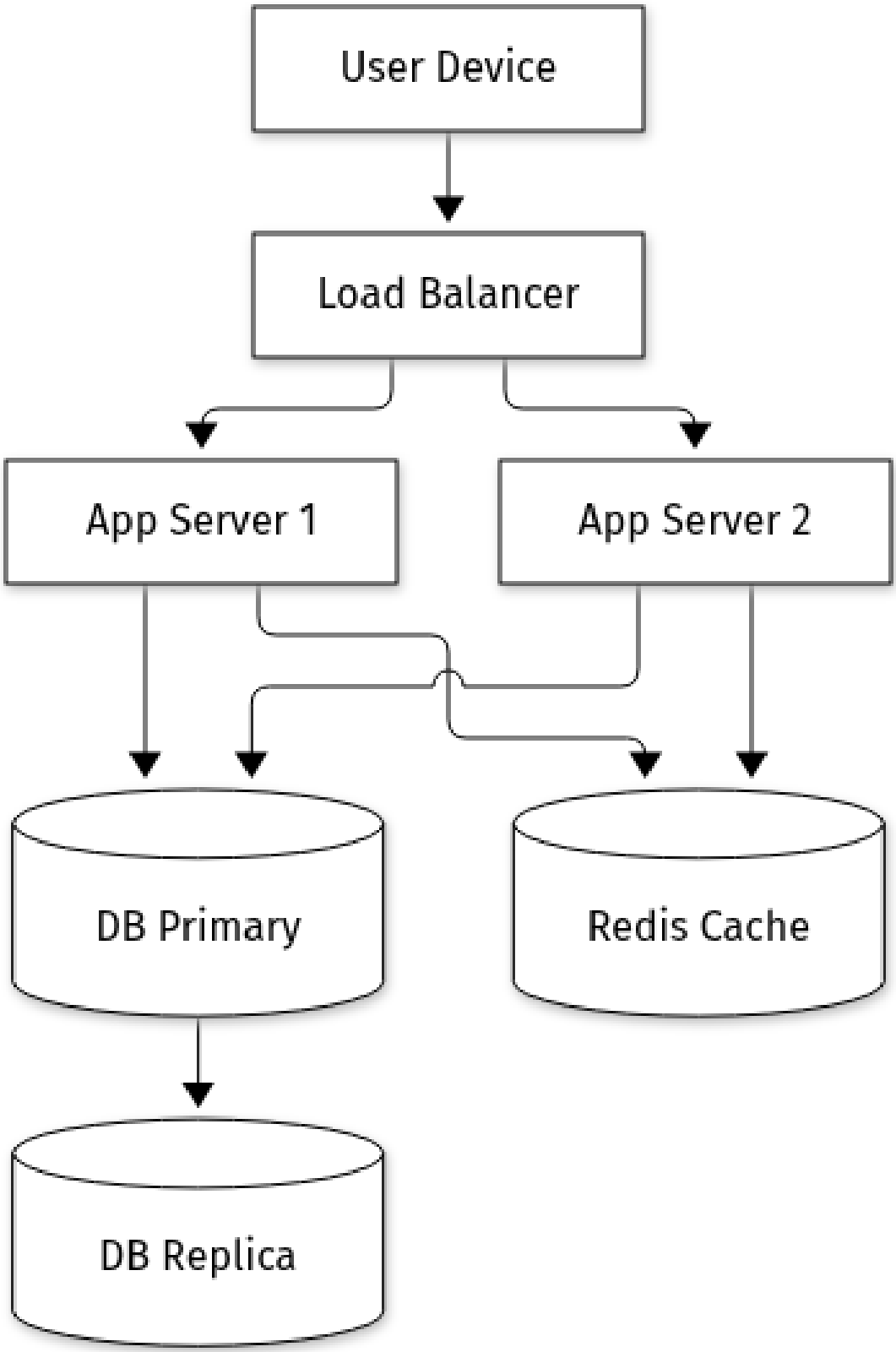
4.12 Package Diagram



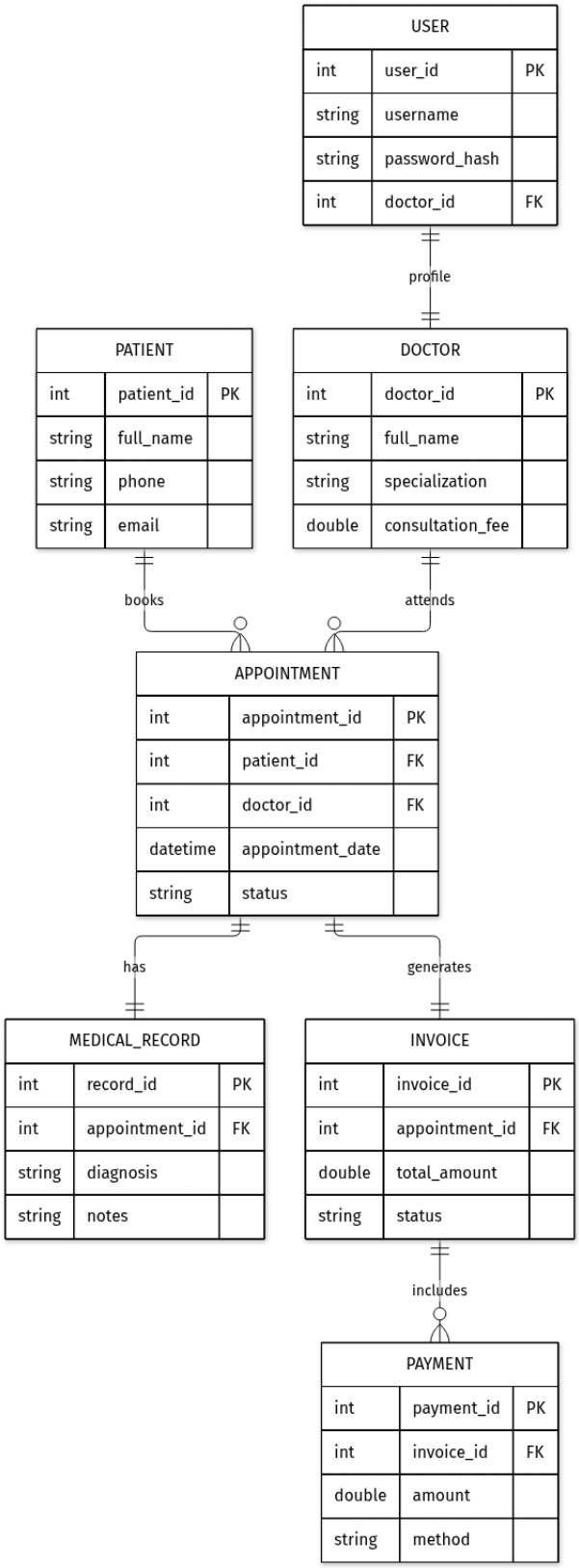
4.13 Component Diagram



4.14 Deployment Diagram



4.15 Entity Relationship Diagram (ERD)



5. Analytical Tables

5.1 CRUD Matrix

Entity	Patient	Doctor	Receptionist	Admin
Patient Record	R,U	R	C,R,U	C,R,U,D
Appointment	C,R,U,D	R,U	C,R,U,D	C,R,U,D
Medical Record	R	C,R,U	-	R
Prescription	R	C,R,U	-	R
Invoice	R	-	C,R,U	C,R,U,D
Payment	C	-	R	C,R,U,D
Schedule	R	C,R,U,D	R	C,R,U,D
User Account	R,U	R,U	R,U	C,R,U,D
Role	-	-	-	C,R,U,D
Reports	-	R	R	C,R

Legend: C = Create, R = Read, U = Update, D = Delete, - = No Access

5.2 Risk Analysis

ID	Risk	Probability	Impact	Severity
R-01	Data breach	Medium	High	High
R-02	Payment downtime	Low	High	Medium
R-03	SMS failure	Medium	Low	Low
R-04	Server overload	Medium	Medium	Medium
R-05	DB corruption	Low	Critical	High
R-06	User resistance	High	Medium	High
R-07	Scope creep	High	Medium	High
R-08	Non-compliance	Low	Critical	High
R-09	Double-booking	Medium	Medium	Medium
R-10	Insider threat	Low	High	Medium

6. User Stories

ID	As a	I want to	So that	Priority
US-01	Patient	Book online	No call	High
US-02	Patient	Receive SMS	Remember	Medium
US-03	Patient	Cancel	Free slot	Medium
US-04	Patient	View history	Track health	High
US-05	Patient	Pay online	Save time	Medium
US-06	Receptionist	Register patient	Onboard	High
US-07	Receptionist	Search phone	Find fast	High
US-08	Receptionist	See availability	No conflicts	High
US-09	Receptionist	Generate invoice	Pay now	High
US-10	Receptionist	Reschedule	Handle changes	Medium
US-11	Doctor	View history	Diagnose	High
US-12	Doctor	Add diagnosis	Document	High
US-13	Doctor	Create prescription	Get meds	High
US-14	Doctor	Manage schedule	Control	Medium
US-15	Doctor	Attach lab results	Full context	Medium
US-16	Admin	Create users	Team access	High
US-17	Admin	Assign roles	Control	High
US-18	Admin	View revenue	Track	Medium
US-19	Admin	View audit logs	Compliance	High
US-20	Admin	Configure settings	Customize	Medium
US-21	Accountant	Export data	Reconcile	Medium
US-22	IT Manager	Monitor system	Uptime	Medium

7. Appendices

7.1 SQL Queries

```
-- Get appointments for a doctor on a specific day
SELECT a.appointment_id, p.full_name, a.appointment_date
FROM APPOINTMENT a
JOIN PATIENT p ON a.patient_id = p.patient_id
WHERE a.doctor_id = 101 ORDER BY a.appointment_date;
```

7.2 Glossary

- BRD: Business Requirements Document
- SRS: Software Requirements Specification
- EMR: Electronic Medical Record
- RBAC: Role-Based Access Control
- NPS: Net Promoter Score
- PK: Primary Key
- FK: Foreign Key

7.3 Tools Used

- Python (Mermaid) — for programmatic diagram generation
- python-docx — for document building
- MS Word (via pywin32) — for PDF conversion
- PostgreSQL — target database
- AWS — target cloud platform

— End of Document —